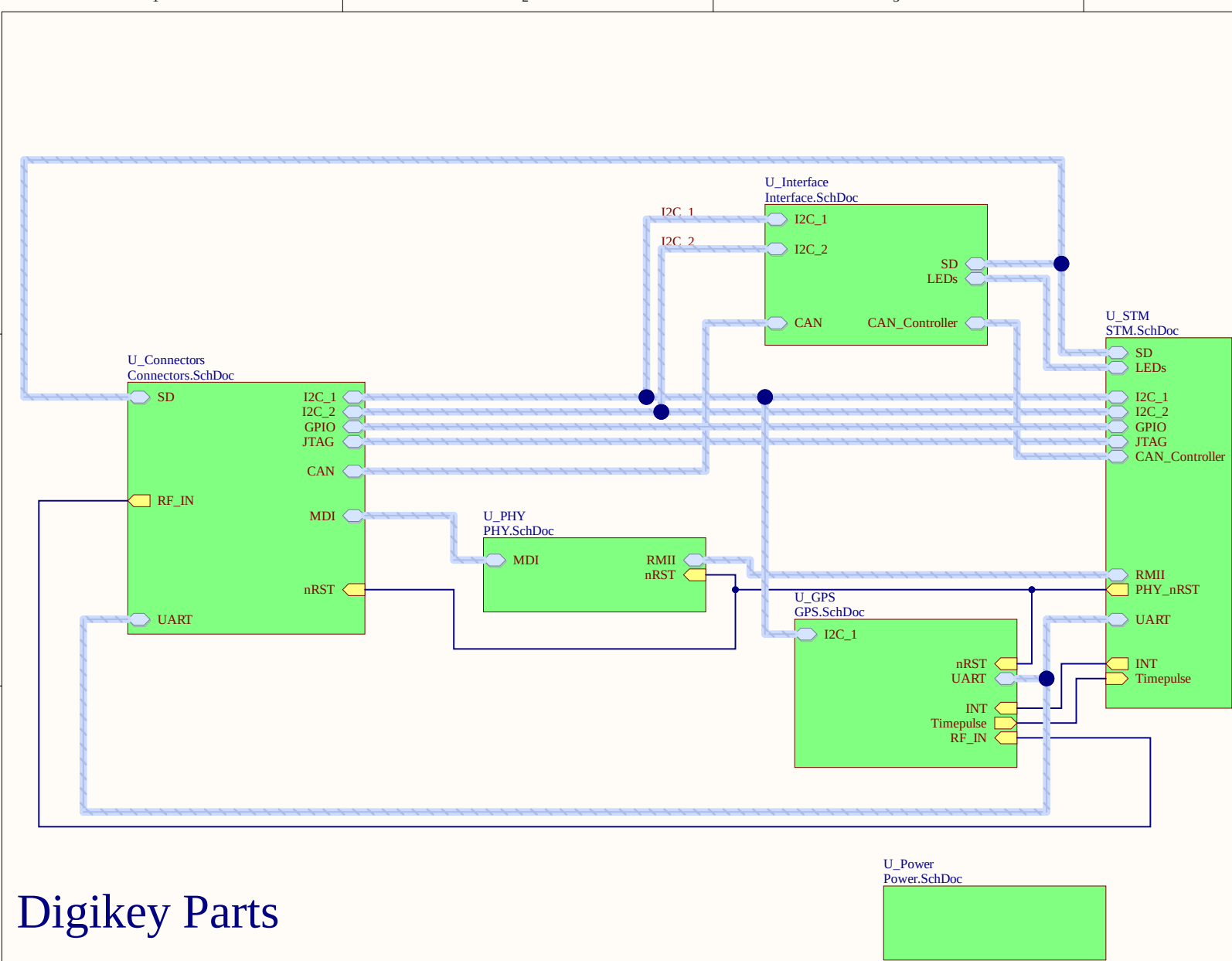




Digikey Parts

Desig.	Part Name	Desig.	Part Name
J1	3 Pin Header	U4	CAN Transciever
J2	34 Pin Ampseal		
J4	2 Pin Header		
J5	2 Pin Header		
J6	2 Pin Header		
J8	2 Pin Header		
J9	2 Pin Header		
J11	8 Pin Header		
BT1	1056TR		

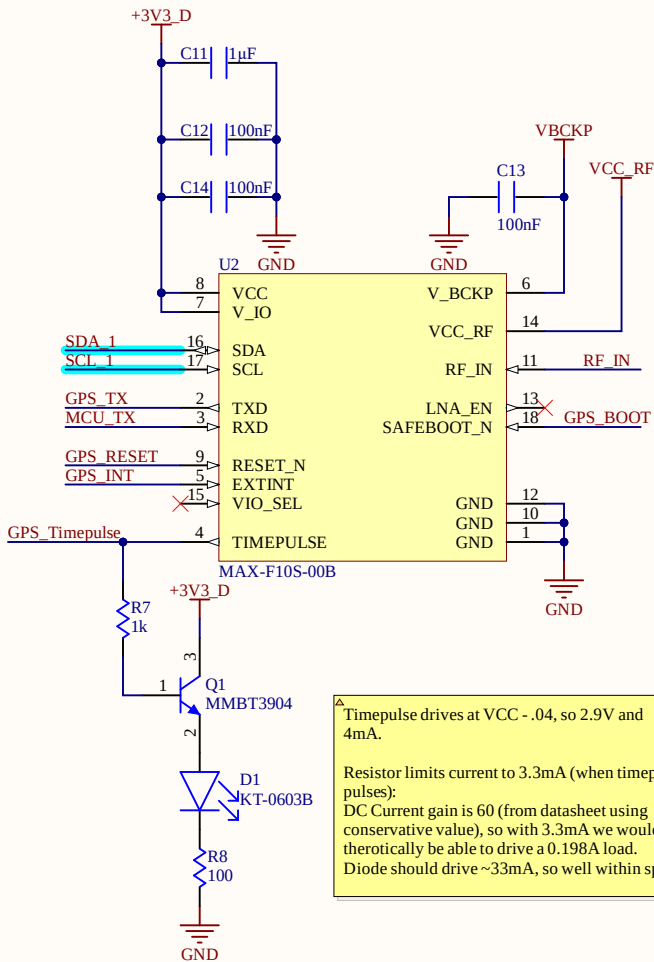
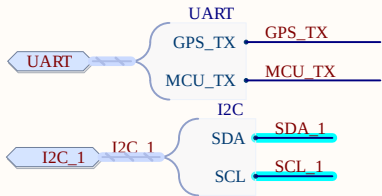
Rev.	Events, Revisions, Details	Date
V1.0	Design completed to requirements.	08/05/2026
	Board Ordered	09/04/2026
	Board Validated & Handed off to Software!	09/25/2026

PROJECT	Digital Board			TITLE		
Last Edit:	10/9/2026			Top Level		
Date:	10/8/2026			CAR	SIZE	REV
Part Num:	ELC-1-DIG-1			MF14	B	1
Drawn By:	Grant Uher			Mines Formula SAE Electrical		
Approved By:				SHEET NUMBER 1 OF 8		



The logo for Mines Formula, featuring the word "MINES" in a large, bold, blue sans-serif font. Below it, the word "FORMULA" is written in a stylized, italicized blue font with a red and blue horizontal swoosh underline. A registered trademark symbol (®) is located to the upper right of the word "FORMULA".





Timepulse drives at VCC - .04, so 2.9V and 4mA.

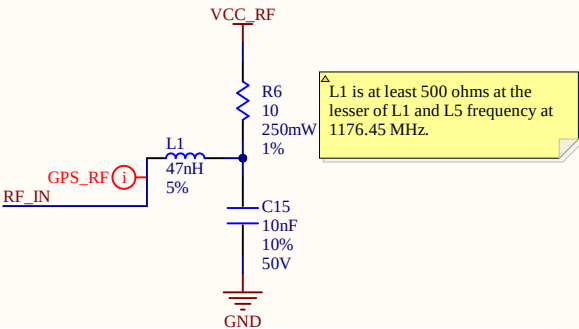
Resistor limits current to 3.3mA (when timepulse pulses):
DC Current gain is 60 (from datasheet using conservative value), so with 3.3mA we would therotically be able to drive a 0.198A load.
Diode should drive ~33mA, so well within specs.

C.1 Standard capacitors

Table 40 presents the recommended capacitor values for MAX-F10S.

Name	Use	Type / Value
C14	RF Bias-T capacitor	10 nF, 10%, 16 V, X7R

Table 40: Standard capacitors

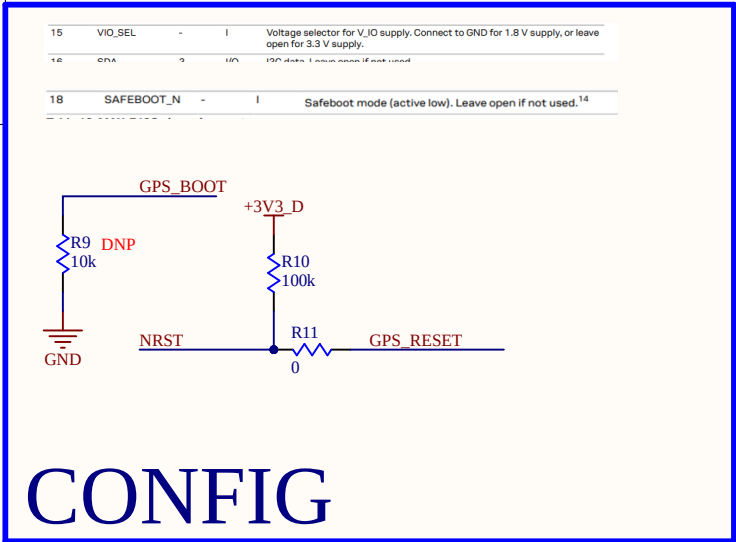
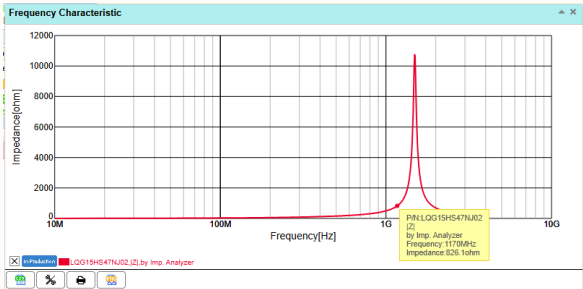


C.3 Inductors

Table 42 presents the recommended inductor values for MAX-F10S.

Name	Use	Type / Value	Recommended component
L4	RF Bias-T inductor	47 nH, 5%	Murata LQG15HS47NJ02 Johanson Technology L-07W series Any other inductor with impedance > 500 Ω at GNSS L1 and L5 frequencies and current rating above 300 mA.

Table 42: Recommended inductors



CONFIG

PROJECT	Digital Board	TITLE	GPS	
Last Edit:	8/19/2026	CAR	MF14	
Date:	10/8/2026	SIZE	B	
Part Num:	ELC-1-DIG-1	REV	1	
Drawn By:	Grant Uher	SHEET NUMBER	3 OF 8	

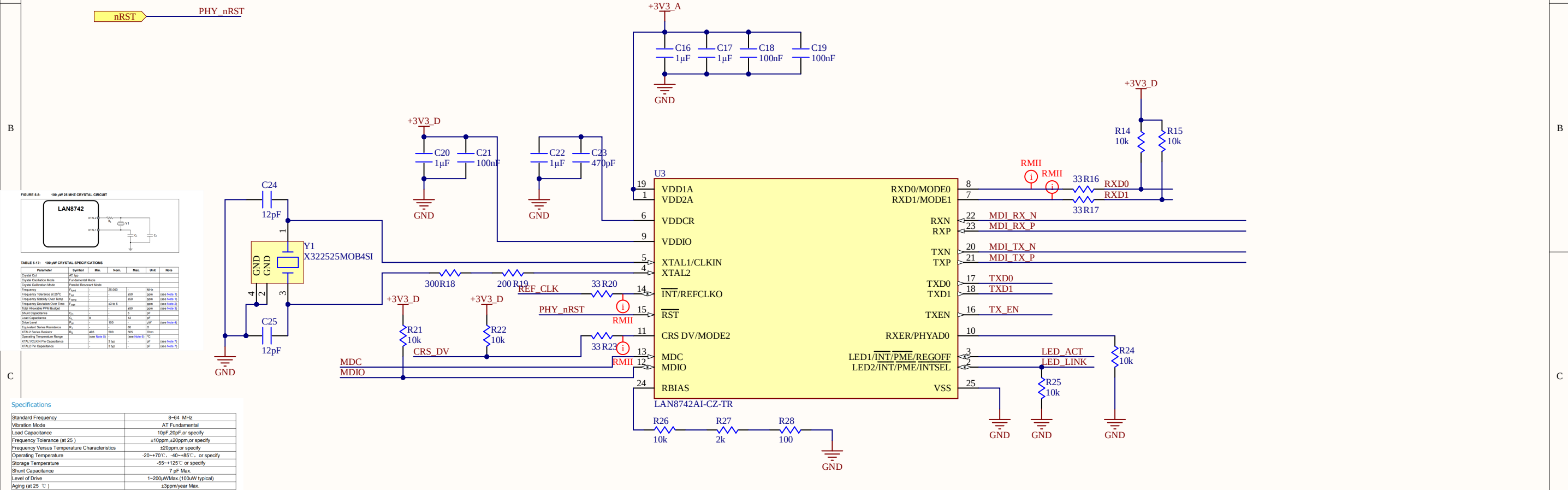


FIGURE 3-15: LED2/nINTSEL POLARITY CONFIGURATION

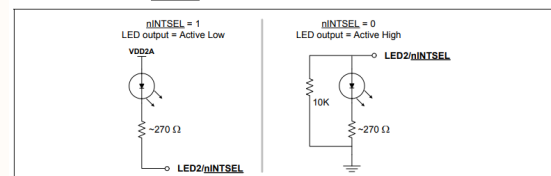
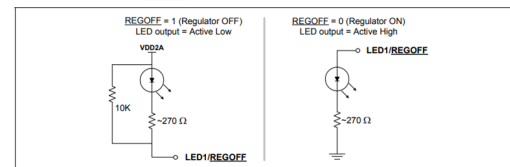
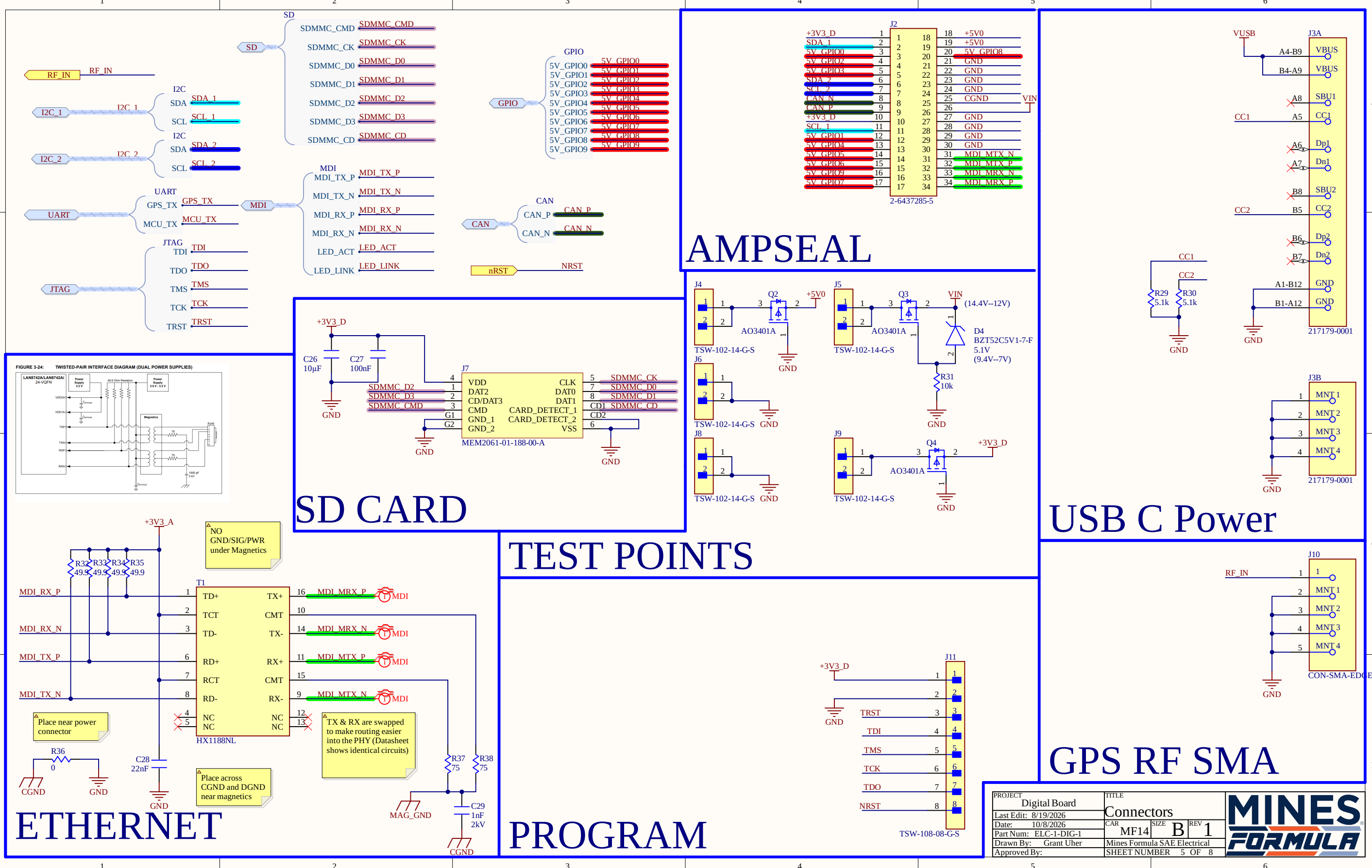


FIGURE 3-14: LED1/REGOFF POLARITY CONFIGURATION



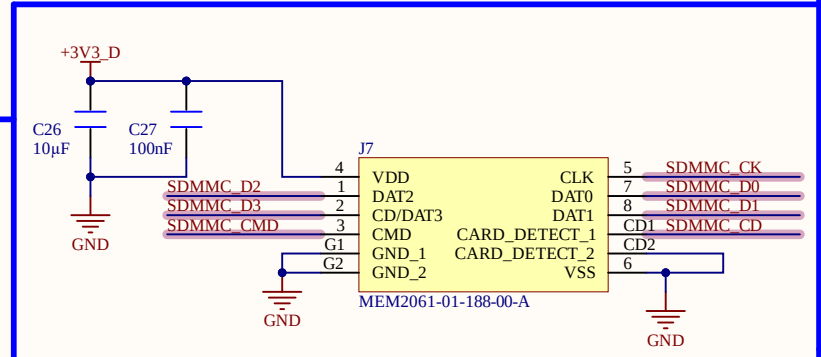
PROJECT	Digital Board			TITLE	Ethernet PHY			
Last Edit:	8/19/2026			CAR		SIZE	B	
Date:	10/8/2026					REV	1	
Part Num:	ELC-1-DIG-1							
Drawn By:	Grant Uher			Mines Formula SAE Electrical				
Approved By:				SHEET NUMBER 4 OF 8				



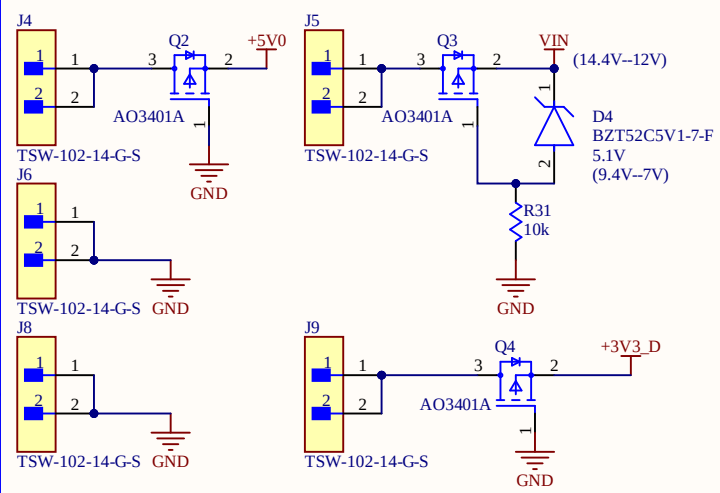
AMPSEAL

+3V3_D	1	18	+5V0
SDA_1	2	19	+5V0
5V_GPIO0	3	20	5V_GPIO8
5V_GPIO2	4	21	GND
5V_GPIO3	5	22	GND
SDA_2	6	23	GND
SCL_2	7	24	GND
CAN_P	8	25	CGND
CAN_N	9	26	
+3V3_D	10	27	GND
SCL_1	11	28	GND
5V_GPIO1	12	29	GND
5V_GPIO4	13	30	GND
5V_GPIO5	14	31	MDI_MTX_N
5V_GPIO6	15	32	MDI_MTX_P
5V_GPIO7	16	33	MDI_MRX_N
5V_GPIO9	17	34	MDI_MRX_P

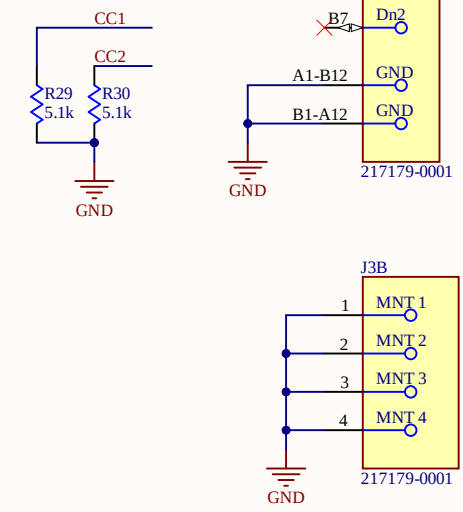
SD CARD



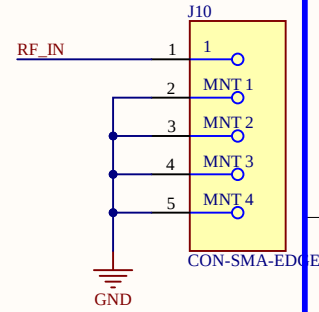
TEST POINTS



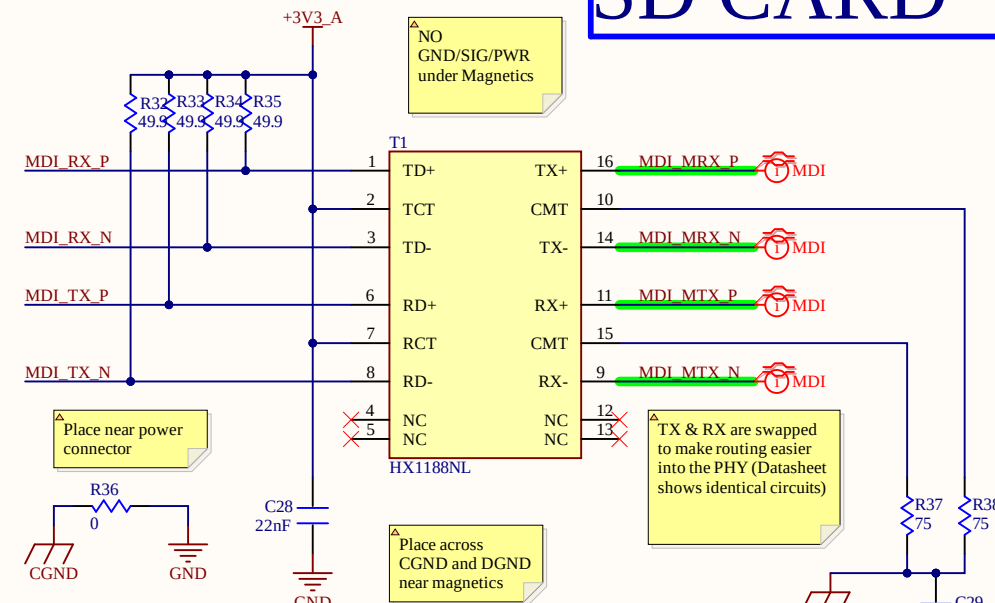
USB C Power



GPS RF SMA



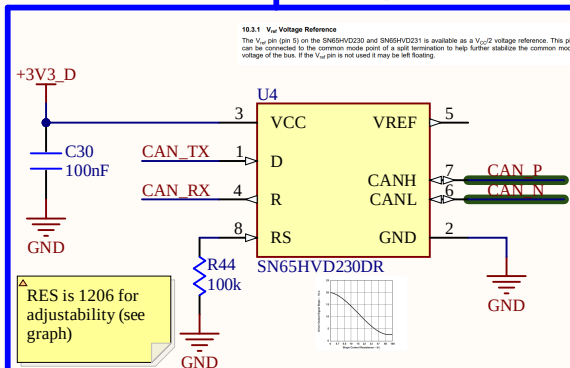
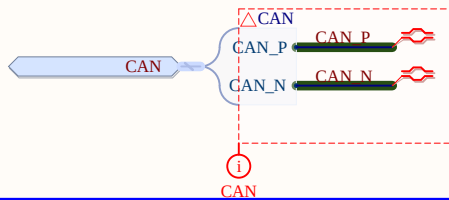
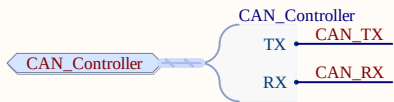
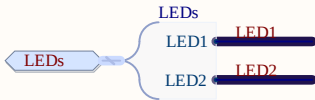
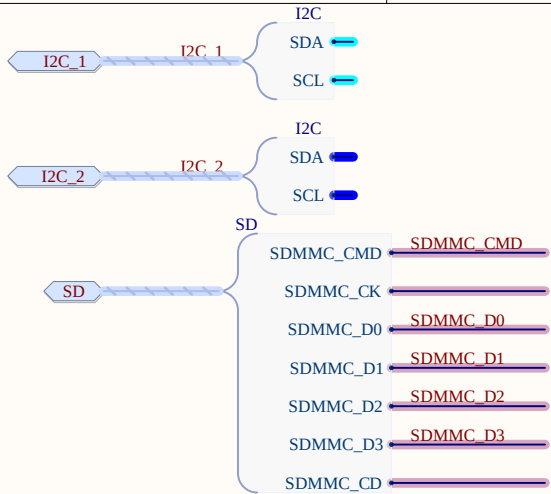
ETHERNET



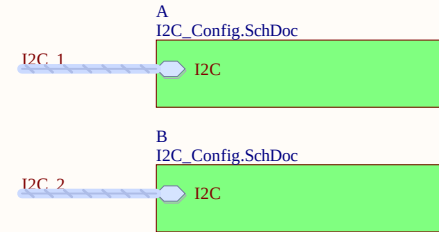
PROGRAM

PROJECT	Digital Board	TITLE	Connectors
Last Edit:	8/19/2026	CAR	MF14
Date:	10/8/2026	SIZE	B
Part Num:	ELC-1-DIG-1	REV	1
Drawn By:	Grant Uher	Mines Formula SAE Electrical	
Approved By:		SHEET NUMBER	5 OF 8

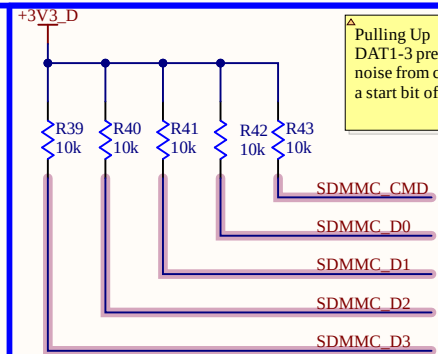




CAN DNP

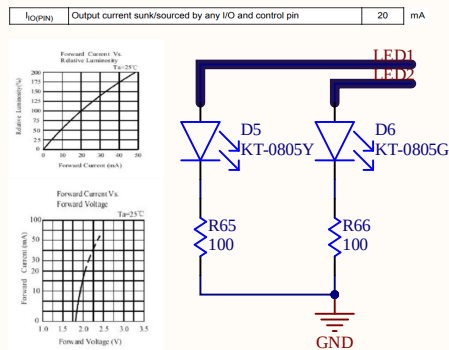


I2C



Pulling Up
DAT1-3 prevents
noise from calling
a start bit of low.

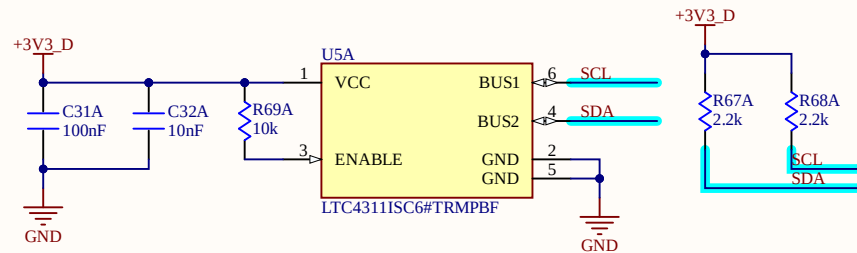
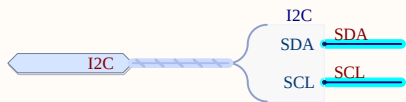
SD Pull Ups



LEDs

PROJECT	Digital Board	TITLE	Interface
Last Edit:	8/19/2026	CAR	MF14
Date:	10/8/2026	SIZE	A
Part Num:	ELC-1-DIG-1	REV	1
Drawn By:	Grant Uher	Mines Formula SAE Electrical	
Approved By:		SHEET NUMBER	6 OF 8

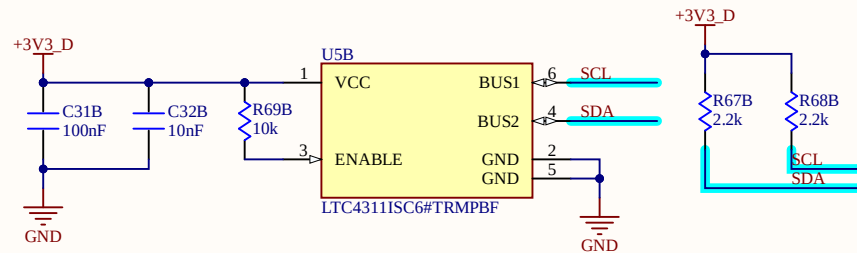
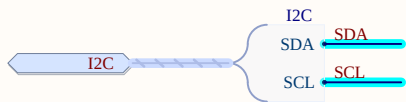
MINES
FORMULA



Place 10nF closer than 100nF caps

PROJECT	TITLE		
Digital Board	I2C Range Extender		
Last Edit: 8/19/2026	CAR	SIZE	REV
Date: 10/8/2026	MF14	A	1
Part Num: ELC-1-DIG-1	Mines Formula SAE Electrical		
Drawn By: Grant Uher	SHEET NUMBER 7 OF 8		
Approved By:			





Place 10nF closer than 100nF caps

PROJECT	Digital Board			TITLE	I2C Range Extender			
Last Edit:	8/19/2026			CAR	MF14	SIZE	A	
Date:	10/8/2026			REV	1			
Part Num:	ELC-1-DIG-1			Mines Formula SAE Electrical				
Drawn By:	Grant Uher			SHEET NUMBER 7 OF 8				
Approved By:								

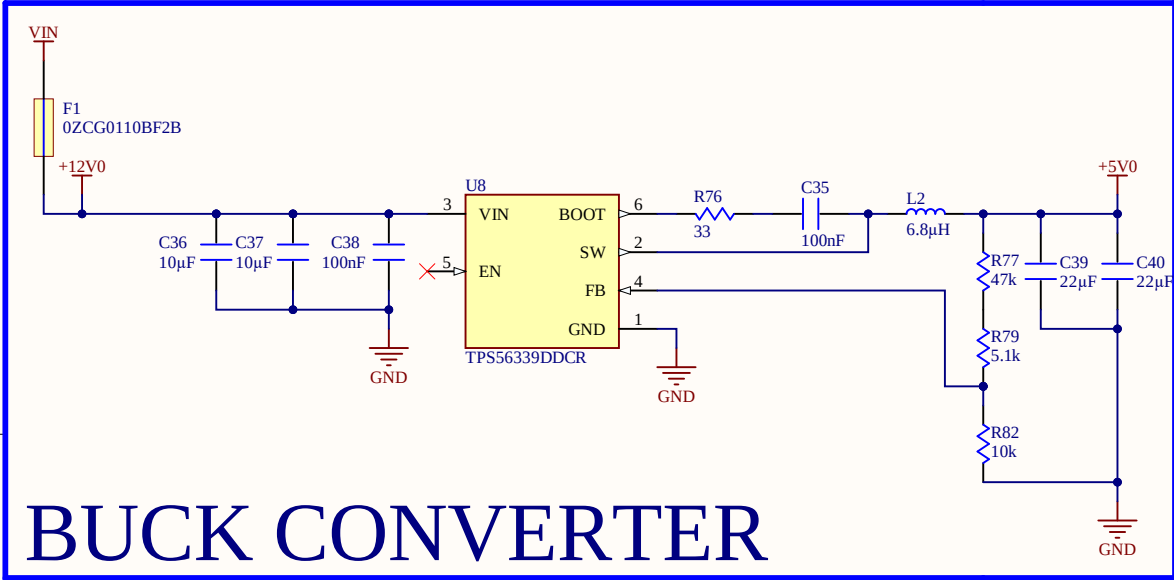


Power Budget:
12V:
LED: 12mA

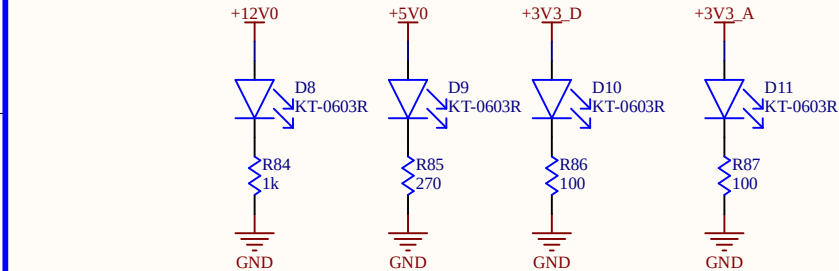
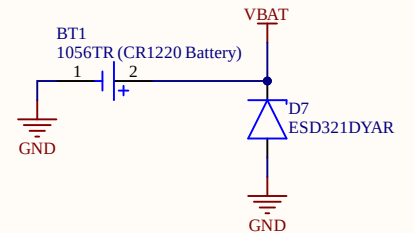
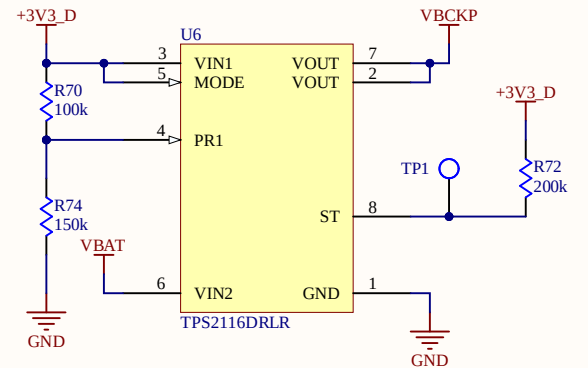
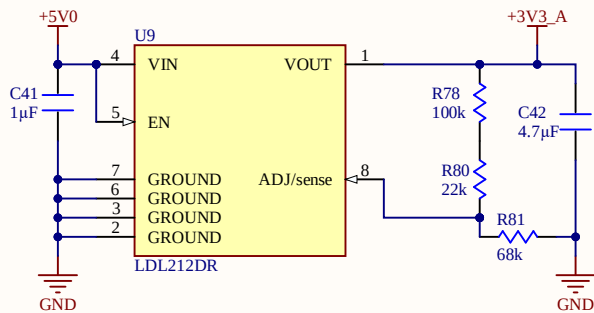
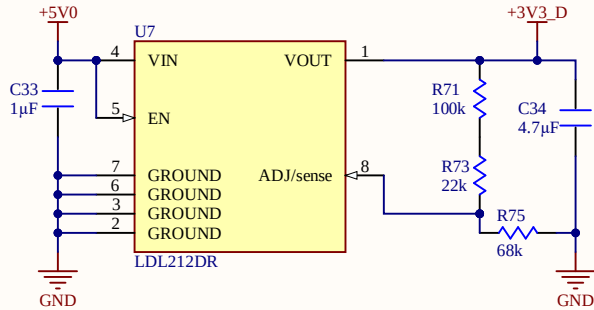
5V:
LED: 18mA

3.3V:
STM32: 120mA
NEO-M9N: 100mA
LAN8742A: 140mA
LEDs x2: 33mA

Total: around 450mA

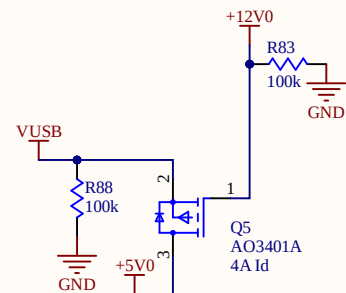


BUCK CONVERTER



LEDs

LDOs



USB Backpower Prot

GPS BACKUP

PROJECT		TITLE			
Digital Board		Power			
Last Edit: 8/19/2026		CAR	SIZE	REV	
Date: 10/8/2026		MF14	B	1	
Part Num: ELC-1-DIG-1		Mines Formula SAE Electrical			
Drawn By: Grant Uher		SHEET NUMBER 8 OF 8			
Approved By:					